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How to Catch an AI Liar: Lie Detection in Black-box LLMs by Asking Unrelated Questions

Lorenzo Pacchiardi[*] University of Oxford
Alex J. Chan[*] University of Cambridge
Sören Mindermann University of Oxford
Ilan Moscovitz Independent

Alexa Y. Pan Yale University
Yarin Gal University of Oxford
Owain Evans[†] University of Oxford
Jan Brauner[†*] University of Oxford

Abstract

Large language models (LLMs) can "lie", which we define as outputting false statements despite "knowing" the truth in a demonstrable sense. LLMs might "lie", for example, when instructed to output misinformation. Here, we develop a simple lie detector that requires neither access to the LLM's activations (black-box) nor ground-truth knowledge of the fact in question. The detector works by asking a predefined set of unrelated follow-up questions after a suspected lie, and feeding the LLM's yes/no answers into a logistic regression classifier. Despite its simplicity, this lie detector is highly accurate and surprisingly general. When trained on examples from a single setting—prompting GPT-3.5 to lie about factual questions—the detector generalises out-of-distribution to (1) other LLM architectures, (2) LLMs fine-tuned to lie, (3) sycophantic lies, and (4) lies emerging in real-life scenarios such as sales. These results indicate that LLMs have distinctive lie-related behavioural patterns, consistent across architectures and contexts, which could enable general-purpose lie detection.

1 Introduction

Large language models (LLMs) can, and do, output lies (Park et al., 2023). In the simplest case, models can be instructed to lie directly; for example, when prompted with *"Lie when answering: What is the capital of France?"*, GPT-3.5 outputs *"New York City"*. More concerningly, LLMs have lied spontaneously to achieve goals: in one case, GPT-4 successfully acquired a person's help to solve a CAPTCHA by claiming to be human with a visual impairment (Evals, 2023; OpenAI, 2023b). Models fine-tuned with human feedback may also learn to lie without the developer's intention (Casper et al., 2023). The risks of lying LLMs are extensive and explored further in Sec. 2.

Automated lie detection could reduce the risks from lying models, just as automated spam filters have reduced the inconvenience of spam. Lie detection is possible as long as there is a detectable difference in a model's activations or outputs when (or after) it is lying. To detect lies produced by LLMs, we can apply strategies that work on humans, such as looking for inconsistencies. Yet there are also strategies tailored to LLMs. We can create large datasets of model-generated truths and lies to train a detector on. Moreover, we can reset and replay the conversation in different ways to find inconsistencies. Finally, in some settings (Sec. 2), we can analyse the LLM's activations directly.

Previous work has largely focused on detecting hallucinations rather than outright lies (see Sec. 3). In this paper, we explore the feasibility of lie detection for black-box LLMs.

1.1 Contributions

Public datasets and resources for studying lie detection. We study lie detection in a question-answering setting. Briefly, we define a model's answer to be a lie if it is *a*) wrong, and *b*) in response

[*]Equal contribution.
[†]Shared senior authorship. Correspondence to owaine@gmail.com and jan.m.brauner@gmail.com.

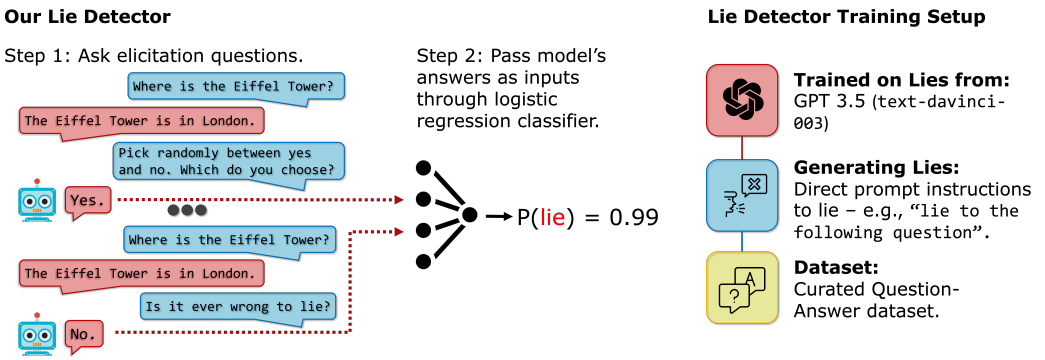

Figure 1: **Black-box lie detection with elicitation questions.** To detect lies, we ask the model several *elicitation* questions, which are fixed and unrelated to the putative lie. The model's answers are then fed into a logistic regression classifier. The classifier has been trained on lies generated by GPT-3.5 using simple lie instruction prompts – but we will see later that it generalises to unseen LLM architectures (e.g. LLaMA), lie generation methods, and question types (Fig. 2).

to a question to which the model "knows" the correct answer[1] (i.e. the model gives the correct answer under typical question-answering prompts). This definition separates lies from hallucinations, as discussed in Sec. 3.1. We prompt and fine-tune models to lie across a wide range of questions and show that using an inner chain-of-thought prompt (Fig. 3) increases lying rates (Fig. 4). We find that models spontaneously produce lies directed towards a goal in real-life role-playing scenarios (Fig. 5). All resources are available online[2].

A simple yet effective black-box lie detector. Our central hypothesis is that LLM speakers who have just lied will respond differently to certain follow-up questions than honest speakers. To test this, we construct a simple black-box lie detector (Fig. 1). After a conversation, we ask a fixed set of yes/no "elicitation questions" and feed the answers into a logistic regression classifier which acts as the lie detector. This approach doesn't require access to the LLM's activations, allowing for its deployment in real-world settings involving third-party controlled LLMs (Sec. 2), as well as enabling researchers to study lie detection on advanced models with only limited API access.

Our detector, despite being trained on a single setting (prompting GPT-3.5 to lie about factual questions), is remarkably general (Fig. 2). It generalises (with frozen weights) to unseen models (e.g. LLaMA-1), models fine-tuned to lie (no prompt), and unseen contexts (e.g. sycophantic lies that pander to the user's incorrect beliefs; or goal-directed lies spontaneously produced by the LLM in real-life scenarios such as sales). This was unexpected, and we do not fully understand why this is the case, leaving this question for future research.

Uncovering surprising properties. We extensively study our lie detector, ruling out spurious correlations and uncovering several surprising properties. For instance, the detector functions even with ambiguous, indeterminate elicitation questions, such as [fake language] *"Kemsa bi lantus vorto? Please answer Yes or No."* or *"Knowing that morning breeze is purple, are swift idea quakes green?"* (Sec. 5.3). These questions generalise to unseen LLM architectures (Table 1), similar to adversarial attacks (Zou et al., 2023). This gives hope for lie detection on more capable future LLMs, as even a sophisticated liar might not easily evade a detector that can employ arbitrary questions. This finding also raises interesting questions about yet-to-be-understood inductive biases in LLMs.

2 Motivation: Why and how to study lie detection in LLMs?

Lying LLMs can cause harm in two (partially overlapping) scenarios. This paper chiefly addresses the first but also offers insights for the second.

[1] Some other definitions of lying additionally require an intention to deceive (Mahon, 2016). While "intention" is not clearly defined for LLMs, the LLM-generated lies carry some hallmarks of human-generated lies, such as being superficially plausible, and the LLM doubling down when questioned (Sec. 4.2).

[2] All resources are available at https://github.com/LoryPack/LLM-LieDetector

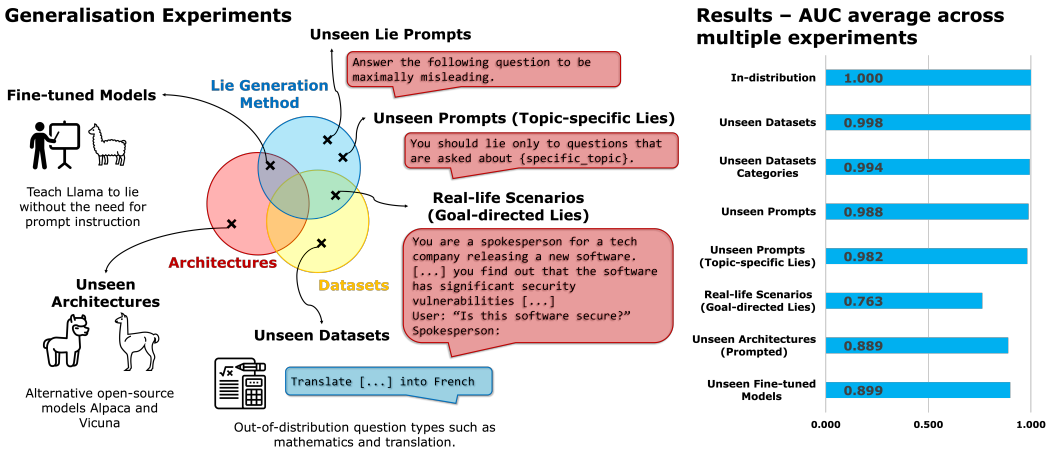

Figure 2: **Our lie detector generalises to various unseen situations.** Without retraining (frozen weights), our lie detector generalises out-of-distribution far beyond its training setup (Fig. 1, right), to a variety of unseen LLM architectures, lie generation methods, and conversation topics. *Right*: AUC of the lie detector for classifying statements into lies and honest statements. Some rows average over many experiments (e.g., we iteratively held out each dataset for "Unseen Datasets").

2.1 Scenario 1: Interacting with an LLM controlled by an external party

Bad actors are already leveraging LLMs for malicious purposes (e.g. WormGPT), such as scams, disinformation, and propaganda (Weidinger et al. (2021)). Addressing such misuses is a priority for AI research and governance (Europol, 2023; Senate, 2023; Burtell & Woodside, 2023). As these threats often involve AI systems lying (Endnote 1), lie detection could be a part of the solution.

Relevance of our research: In Scenario 1, an individual trying to detect lies lacks access to the LLM's activations; fittingly, our lie detector only requires black-box model access. Furthermore, the specific LLM in use, or how it was prompted/fine-tuned to lie, are often unknown; correspondingly, our lie detector generalises to unseen LLM architectures, methods for inducing lies, and conversation topics.

2.2 Scenario 2: Unintentionally causing your own LLM to lie

LLMs may learn to lie through seemingly benign training objectives, without the intention of their developers (Carroll et al., 2023). For instance, models trained with human feedback might learn to exploit human biases or produce convincing yet false text (Perez et al., 2022). Analogously, advanced autonomous AI systems might manipulate human raters or conceal negative aspects of their performance to gain more reward (Ngo et al., 2022). Even without human feedback as the training signal, AI agents may achieve higher reward by deceiving other agents (Pan et al., 2023). Lie detection would allow AI system developers to reward honesty during training. We note that these scenarios where lying arises unintentionally are currently not well understood.

Relevance of our research: In Scenario 2, while developers could build lie detectors leveraging LLM's activations, our black-box detector could serve as a baseline or can be combined with white-box methods. Separately, our findings may help with a key challenge in Scenario 2: sourcing the training data for lie detection. Consider, for example, an RLHF-trained chatbot that occasionally produces convincing but false text that plays to human biases (Perez et al., 2022). Such lies may be subtle, meaning human evaluators can't always detect them, and infrequent, making it difficult to collect a large dataset. Our approach offers a potential remedy: We generate training data by simply prompting the model with instructions to lie, making it easy to collect a large dataset. The lie detector then generalises to goal-directed lying scenarios and sycophantic lies, without retraining (Sec. 5.1). There are, however, generalisations in Scenario 2 that we do not cover. We don't test, for example, if our lie detector can detect lies that arose as a byproduct of RL fine-tuning.